

EE595H: Quiz

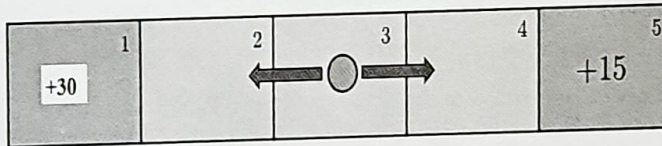
Total Questions: 2, Total Marks: 35, Time: 60 Mins

1. An absent-minded professor has three locations: Home, Office, and the Faculty Club. Each day, he commutes from Home $\rightarrow$ Office, then Office $\rightarrow$ Faculty Club, and finally Faculty Club $\rightarrow$ Home. He has a total of 3 umbrellas. If it rains (probability p) and he has an umbrella at his current location, he takes it. If it doesn't rain, he leaves all umbrellas behind. Rain events for each commute are independent.

Find the steady-state probability that he gets wet on his commute from the Faculty Club back Home.

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2. Consider a simplified game problem defined over the grid shown below. The figure shows two terminal states $s \in \{1, 5\}$ with rewards $r(1) = +30$, $r(5) = +15$, and three additional states $s \in \{2, 3, 4\}$ constant reward $r(s) = -1$. The terminal squares lead to the EXIT state where the game stops. In the center states, the agent can select equally between two actions: $a \in \{\text{RIGHT}, \text{LEFT}\}$. Assume that whenever the action RIGHT is chosen, there is a 25% chance that the agent will move left. There is no randomness with the action LEFT. The discount factor is set to $\gamma = 0.5$.



- (a) Determine the state transition probability matrix, P^π .
- (b) Determine the one-step expected reward vector, r^π .
- (c) Determine the state values, $v^\pi(s)$, under the assumed policy.

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Total: 25

EE595H (Stochastic Models): Half Semester Examination

Total Questions: 2, Total Marks: 35, Time: 120 Mins

1. A die is rolled repeatedly. Which of the following are Markov chains? For those that are, derive the transition matrix.

- (a) The largest number X_n shown up to the n th roll.
- (b) The number N_n of sixes in n rolls.
- (c) At time r , the time C_r since the most recent six.
- (d) At time r , the time B_r until the next six.

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Total: 23

2. Consider a Markov decision process (MDP) with three states

$$S = \{s_1, s_2, s_3\}$$

and two actions

$$A = \{a_1, a_2\}.$$

The discount factor is $\gamma = 0.9$. The transition probabilities and immediate rewards are given in Table 1.

Table 1: Transition Probabilities and Rewards

State	Action	Next State	Transition Prob.	Reward
s_1	a_1	s_1	0.2	3
s_1	a_1	s_2	0.6	2
s_1	a_1	s_3	0.2	0
s_1	a_2	s_2	0.5	4
s_1	a_2	s_3	0.5	2
s_2	a_1	s_1	0.4	1
s_2	a_1	s_3	0.6	3
s_2	a_2	s_2	0.3	2
s_2	a_2	s_3	0.7	6
s_3	a_1	s_1	0.3	-2
s_3	a_1	s_3	0.7	1
s_3	a_2	s_3	1.0	0

Initialize the value function as

$$v_0(s_1) = v_0(s_2) = v_0(s_3) = 0.$$

Using Optimal Value Iteration,

- (a) Compute the value functions after the first iteration.
- (b) Compute the value functions after the second iteration.
- (c) Determine the greedy deterministic policy after the second iteration.

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Total: 12

$$V(s_1) = 3$$

$$V(s_2) = 4.3$$

$$V(s_3) = 0.1$$